failures and important edge cases. For each case, write down what success means before running the test. Some checks can be objective: Was the correct record updated? Did the calculation match? Did the code pass its tests? Others need a simple human rubric: Was the answer useful? Did it acknowledge uncertainty? Would a reasonable user know what to do next?

Run the test set more than once. Generative AI can respond differently even with the same input, so one successful run does not tell you whether the system is reliable or simply got lucky. It also helps to maintain two test sets: one for new and difficult tasks and another for tasks the system already handles well. The second is a safety net whenever you change the model, prompt, data source or tools.

Then save every meaningful failure as a future test case. Over time, the test set becomes a record of what the product has learned not to break.

Q Gen AI can produce many valid answers. What should we actually measure instead of "accuracy"?

A Spurthi: There is no universal metric. I would start with the most basic question: did it actually help the user complete the task? Then: was the answer correct and useful, and does the system perform consistently rather than occasionally? What happens when it gets something wrong? Is the mistake obvious, and can it be corrected? Is it practical? An excellent answer that is slow or expensive or needs extensive human checks may not be of value.

The balance changes with the application. In brainstorming, variety may be more valuable than consistency. When extracting information from invoices, correctness and consistency matter much more. And if an agent can issue refunds or approve payments, its permissions, safeguards, and ability to reverse an error become as important as the quality of its response.



This is why I would resist reducing an AI feature to "92% accurate." 92% of what, in which cases, and with what consequences for the other 8 percent?

Q Where does human evaluation matter? Are there situations where automated evaluation can't tell us whether a system is good enough?

A Spurthi: Human evaluation matters whenever quality depends on context, judgment, or consequences that are difficult to encode in a rule. A program can check whether a refund was issued, but it can't decide whether the explanation was good for a distressed customer.

If there's a clear right answer or system state to check, yes/no tests are the best place to start. For open-ended outputs, you can use AI models as graders, but they bring their quirks. Studies on "LLM-as-a-judge" show that strong model judges agree with humans, yet they can be swayed by where and how long an answer appears, and how much their own style is matched.

It isn't humans versus automation. Use code for facts and state changes that can be verified directly. Use model-based graders for scale when the rubric is clear. Then calibrate those graders against people, preferably domain experts when the stakes require it, and review disagreements regularly.

Seven questions before you trust an AI tool

1. **Has it been tested on my kind of task?** Not just benchmarks, but examples like your actual work: specialized language, messy data, multi-step workflows.
2. **Can I verify the important output?** A source, a calculation, a test result or a transaction record: an independent way to check.
3. **Does it work consistently?** Try the same task more than once, with slightly different wording.
4. **What happens when it is wrong?** A trustworthy product fails visibly, avoids pretending certainty, and makes it easy to correct, escalate, or say "I don't know."
5. **Can a person intervene or reverse the action?** This matters most when the system can spend money, change records, contact customers, or affect someone's rights.
6. **Is the benefit worth the checking, delay, and cost?** If people spend more time verifying than they save, it may not be worth using.
7. **How serious would one undetected failure be?** The more serious the impact, the stronger the evidence, safeguards, and human oversight you should demand.

"Sounding intelligent is not the same as being reliable," says Tallam. "In some cases, fluent answers actually make serious mistakes harder to spot. AI is really 'working' when it improves a real outcome in the way people actually use it, at a level of error, cost, and risk you can accept."